

United Kingdom Competition and Markets Authority

The CMA's Data, Technology and Analytics unit (DaTA unit) – summary of core projects

On 14 June 2022 the CMA published a discussion paper written by Stefan Hunt, *The technology-led transformation of competition and consumer agencies: The Competition and Markets Authority's experience*,²⁰ which provides more information on the formation of the DaTA unit and the value it has brought to the CMA.

The following note sets out and summarises the core projects which the DaTA unit has undertaken since its inception in 2019.

Digital advertising

The DaTA unit advised on the digital advertising market study which ran for 12 months from July 2019. The DaTA unit helped the case team by:

- Understanding the technology and systems that platforms and adtech providers use, including understanding the role of data in digital advertising and the impacts on competition, privacy and efficiency; and assessing the technical feasibility and effectiveness of potential data-related remedies (access, sharing, controls/restrictions).
- Providing data science and engineering support, including the heavy lifting of designing, coordinating, transferring and analysing 4TB+ worth of datasets from platforms.
- Providing detailed choice architecture analysis to refine interim report proposals.
- Providing support for information requests (e.g., for user testing) to firms, analysis of the responses and development of testing/monitoring proposals for the final report.

Mobile ecosystems market study

In the mobile ecosystems market study, the DaTA unit provided analysis on large data sets from the main Parties as well as insight into mobile ads and Android and Google mobile services. Our behavioural scientists also worked on this study, which covered the core elements of operating systems, app stores, and web browsers on Apple iOS and Google Android. The team analysed how consumers might react to the use of different choice architecture practices on mobile devices and how this can affect competition. This includes evaluating the evidence submitted by stakeholders and academic literature on psychological mechanisms. The team also advised on the design of potential remedies to address the concerns identified.

²⁰ Stefan Hunt, *The Technology-Led Transformation of Competition and Consumer Agencies: The Competition And Markets Authority's Experience*, CMA (Jun. 14, 2022) <https://perma.cc/SNH3-28V8>.

Google Privacy Sandbox

The CMA became aware of Google's proposed phasing out of third-party cookies on Chrome, its web browser, and introduction of new technologies that would support targeted advertising while preventing cross-site tracking: Google's Privacy Sandbox. The CMA had a concern that the proposed changes could undermine the ability of publishers to generate revenue through advertising, undermine competition in digital advertising and allow Google to entrench its market power as both a publisher providing advertising inventory and a provider of adtech services. In January 2021, the CMA launched an antitrust case in advance of the changes, working closely with the UK Information Commissioner's Office. Understanding the implications of the changes for the many parties affected has required getting deep into the technologies, the DaTA unit has attended key meetings with the parties and driven the CMA's understanding of the technology. The CMA accepted commitments from Google setting out how the technology will be developed, with the CMA having an ongoing monitoring role.

Meta's gathering and use of advertising data

The DaTA unit advised on a CMA case involving Meta's gathering and use of advertising data for its online classified ads and online dating services. It quickly became clear that understanding the use of data was going to require grappling with how Meta's algorithmic systems function, and so we deployed a data scientist with many years of experience in using machine learning algorithms, including within tech firms. The data scientist made a large difference by enabling the case team to pursue important lines of inquiry that they otherwise would not have.

Online reviews

The DaTA unit has supported the CMA's consumer team with their series of online reviews cases. The first phase of the work focused on the platforms where fake and misleading reviews are traded. This involved meeting, as part of the case team, with eBay and Facebook on their proposals to detect and remove the trading of reviews. The second phase of the work involved the identification of which review businesses to prioritise for in-depth investigation and the subsequent launch of investigations into Google's and Amazon's approaches to identifying and moderating fake and misleading online reviews. As fully embedded case team members, the DaTA unit wrote substantial parts of the detailed requests for information, including requests for considerable data and details of the algorithmic systems that Google and Amazon have built.

The DaTA unit has conducted extensive web scraping on online reviews and used techniques to detect suspicious patterns indicative of fake and misleading reviews. To decide which companies to investigate in-depth, we collected evidence from several different platforms. We used techniques both to detect suspicious patterns ourselves and to ensure that we had a thorough grip on the methods, so in any subsequent investigations we could assess the advantages and drawbacks of firms' approaches.

Case support for mergers

The DaTA unit has provided direct input to mergers cases and performs three different roles as part of a mergers case team:

- First, we develop specific data or technology-focused theories of harm. Two examples are the Amazon/ Deliveroo case and the Meta/Giphy case.
- Second, we aid the case team in understanding the overall market or specific aspects of products in cases where the DaTA unit has preexisting knowledge of the merging sector. Cases here include Google/Looker and Salesforce/Tableau where cloud technology and the variety of different services on the cloud was a major focus.
- Third, we help assess technical remedies, particularly relevant for intellectual property remedies. One example is Tobii/Smartbox where a remedy was proposed that would make the code from Tobii for their products available on an open-source basis.

CMA's Covid Taskforce

The context for our work in this space was that markets were changing rapidly given the shock of lockdown and the introduction of Covid regulations, leading to price spikes and the unexpected cancellation of travel and pre-booked events.

The DaTA Unit built a CMA webform and data pipeline, which was required to be able to receive, process, and analyse over 100,000 complaints from the public about consumer-related issues since the beginning of the Covid-19 pandemic. Data scientists systematically corrected business names through deduplication and matching and also determined the relevant market or sector to which a complaint related using machine learning techniques.

Without the work of the DaTA Unit the CMA would not have had its own real-time complaints information from which it was able to identify issues and would have required a larger number of staff to work on processing information coming through the existing public-facing CMA channels.

Document review

Our Evidence Submission Portal (ESP) is now used for large document submissions to the CMA. The ESP can take in millions of documents, check that they are in the right format, and process them.

ESP exemplifies a situation where there was no existing solution to meet our requirements, and our data engineers and eDiscovery specialists built a product that integrates well with our commercial eDiscovery tool, Nuix Discover. We have reduced the amount of time from submission of documents by the parties to when the merger case team can review the documents from 4 days down to 1 day, which can make a big difference when under time pressure.

In addition, using natural language processing techniques, we have started deploying new capabilities into ESP that provide additional insight on individual documents, helping to speed up the evidence review process. We have deployed three data science enrichments to Nuix. The first is a 'keywords enrichment,' which extracts the most representative words for each document, allowing reviewers to better understand the likely content. The second indicates how likely it is that a document is about competition-related topics (e.g., competitors, prices, market structure, etc.). The third indicates whether a document talks about activity that is within the UK.

Companies Network Project

The DaTA unit is creating a data pipeline which takes records from all registered limited companies from Companies House, the UK's registrar. These data are frequently needed by the CMA for many reasons, including i) getting intelligence on companies with respect to suspected cartel activity, ii) understanding ownership structures in markets, and iii) understanding the state of markets, especially concentration and profitability levels. Ordinarily CMA staff navigate the publicly available search tool and download the data by hand. But this is time-consuming, and the manual nature of the process could lead to errors. The pipeline will regularly take in all the data, clean it, deduplicate it, and make it available in an easy-to-use tool, designed for the CMA's needs (with the pipeline also available more widely). This is a significant engineering challenge, especially as the dataset is reasonably large

The payday lending market investigation

This market investigation concluded by implementing a remedy that lenders needed to put a link to a price comparison website on their webpage. The DaTA unit created code to scrape company websites to check that there was a link, writing to any non-compliant firms and telling them to comply.

DaTA unit's LEDA platform

The DaTA unit uses its own platform LEDA for creating a data lake (a system for storing and accessing raw data) and providing analysts with access to the data lake and to cloud computing services. Our data scientists and engineers need flexible access to coding packages, version control of all code and data (using GitHub) so that we can reproduce all analysis, and the ability to build and control information security and data protection. Working closely with our IT department colleagues, we developed a data infrastructure capable of ingesting, curating, and processing sensitive data at the scale required of a leading competition authority and with modern advanced functionality. In the last three years, we have acquired over 160TB of data across over 130 million objects stored in our cloud-based data lake, at minimal cost and risk. This is important for our quantitative teams, who have excellent alternative employment opportunities and really care about quality of the 'tech stack': they have the tools they need for the job, like the platform, and feel empowered.

Autorenewals cases

The DaTA unit's Behavioural Hub has been working with the CMA's consumer team on a series of cases focusing on autorenewal contracts for McAfee and Norton antivirus products and gaming subscriptions for Microsoft Xbox, Sony, and Nintendo. Consumer behaviour was central to the investigation. The DaTA unit analysed firms' interactions with consumers – their emails, websites, notifications – and the precise elements of the design. Much is known about the effects of this design – defaults, ease of exit, comprehension of contract terms, provision of risk information – and the behavioural insight advisers built up very detailed evidence on the impact of these elements. The team advised on data collection, including what data and information could be requested from firms and how it could be used to support legal arguments. This included advising on how changes introduced by firms could be used to assess the impact on consumers using quasi-experimental methods. As the cases progressed to remedies, the team members were active in assessing the likely impact of the remedies and so what the CMA should accept in terms of undertakings.

Algorithms programme

In January 2021, the DaTA unit published new research on algorithms, showing how they can reduce competition in digital markets and harm consumers if they are misused. The paper aimed to lay out the territory that the CMA would need to cover over the next 5 to 10 years. There was subsequently a panel event which discussed the key issues in more detail.

The DaTA unit has also conducted additional research on algorithmic harms and algorithmic auditing working with other UK regulatory agencies and anticipates publishing a paper later in 2022 on the CMA's practical lessons from cases involving algorithms.

Online choice architecture (OCA) programme

The DaTA unit's Behavioural Hub published a discussion paper and an extensive evidence review in April 2022. The aim of the research was seeking to lay out a roadmap of an increasingly important area. OCA featured prominently in the Furman report and the Stigler Centre report on digital markets. And OCA is currently an important part of many digital cases, e.g., browser defaults on Apple devices, app stores in mobile ecosystems, or privacy settings within Google's Privacy Sandbox.

The first paper outlines the harms that can arise (while noting that OCA can be and often is hugely beneficial) and includes a taxonomy of 21 concerning practices that agencies need to be aware of and alert to. The second paper is a long and detailed review of the practices and cross-cutting themes, which is intended to be a reference document for agencies, providing a trove of evidence to support future cases.